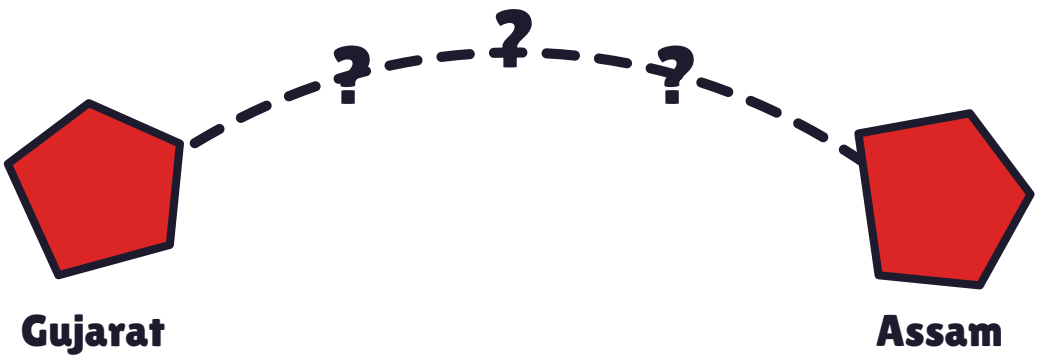
SARHAD

A one-minute daily puzzle about India's borders

सरहद means "border". Build task for the Full Stack Engineer role at Kenpath Labs.



Play it
sarhad-teal.vercel.app
open on a phone

Code
github.com/nitya2405/sarhad
README has setup

By
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19 October 2026

01 What was asked

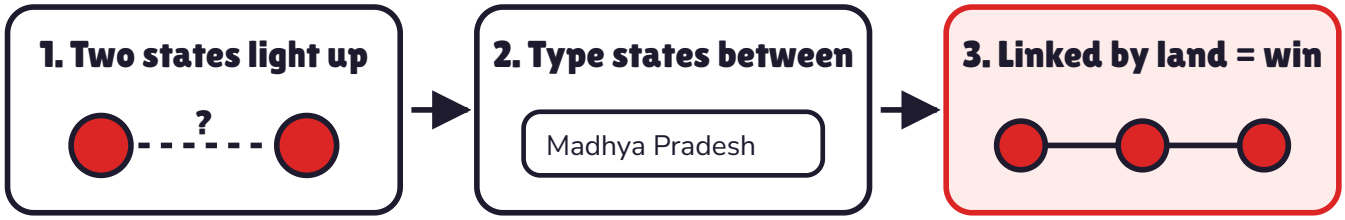
Build a small, responsive web app that people would want to open every day: they do a short task and get rewarded for it, with the retention mechanics that fit the idea.

Kenpath's note	What it means for the design
Most users are on modest Android phones and mobile data	Small download, fast first screen, works on a slow network, nothing heavy to store.
Many read English as a second language	Few words, short sentences, pictures over paragraphs, forgiving input.
What they earn is worth real money to them	Points must be fair, impossible to fake, never lost or doubled, and the rules must be clear.

Deliverables: a live link, a repository with a README, and an explainer. This document is the explainer.

02 What I made

Sarhad shows you two Indian states. You type the states in between until they join by land. Any route counts, and the faster you finish, the more points you get.

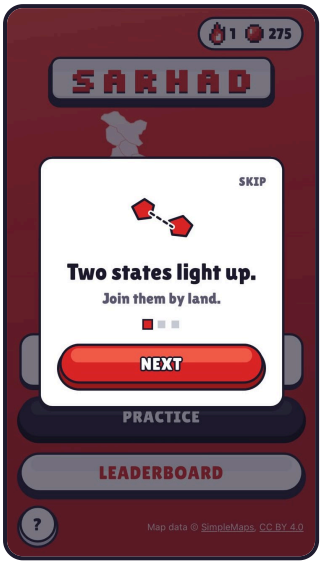


Why this idea

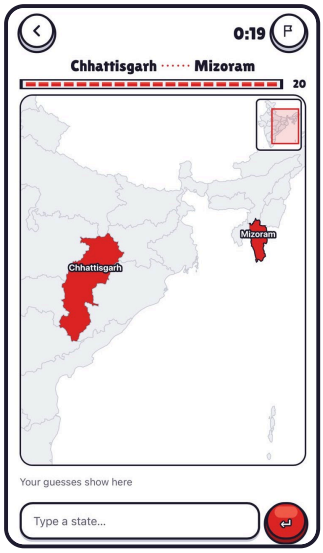
- **One minute a day.** A round takes 20–60 seconds for most players, short enough to fit a daily habit.
- **Everyone already knows the pieces.** State names are familiar to every Indian player, so there is nothing new to learn before playing.
- **Any route wins.** The game never says a correct answer is wrong just because it isn't the stored one. That matters when points are worth something.
- **You learn something even when you lose.** The result shows one shortest route, so the map sticks a little more each day.

03 Simple, light and fun

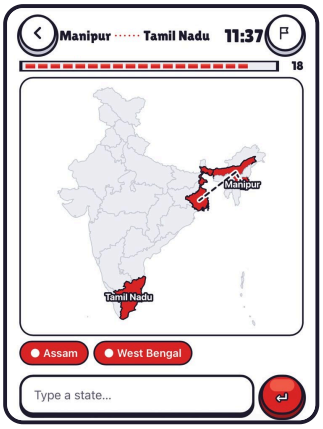
Every screen does one job, uses as few words as possible, and answers every tap straight away.



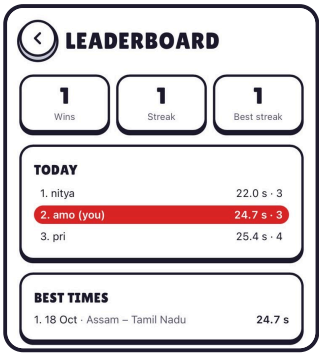
How to play: 3 cards, 6 words each



Game: map zooms to the two states



Linked states turn red and join



Leaderboard and stats

Choice	Why it keeps the game easy
No sign-up	Pick a name and play. A silent anonymous account is created in the background.
Suggestions without spoilers	A suggestion appears only once a name is nearly typed. Typos get one "Did you mean" chip. Abbreviations (MP, UP) and old names (Orissa) work.
Colour plus shape	Linked guesses turn red with a filled dot; unlinked ones are grey and hatched with an empty dot, so colour is never the only signal.
Instant feedback	Short synthesised sounds for start, linked, not linked, wrong input and win, with a speaker toggle. Pop animations respect "reduce motion".
Big, chunky controls	Buttons are at least 48 px tall with a thick outline, so they are easy to hit with a thumb.

04 Why people come back

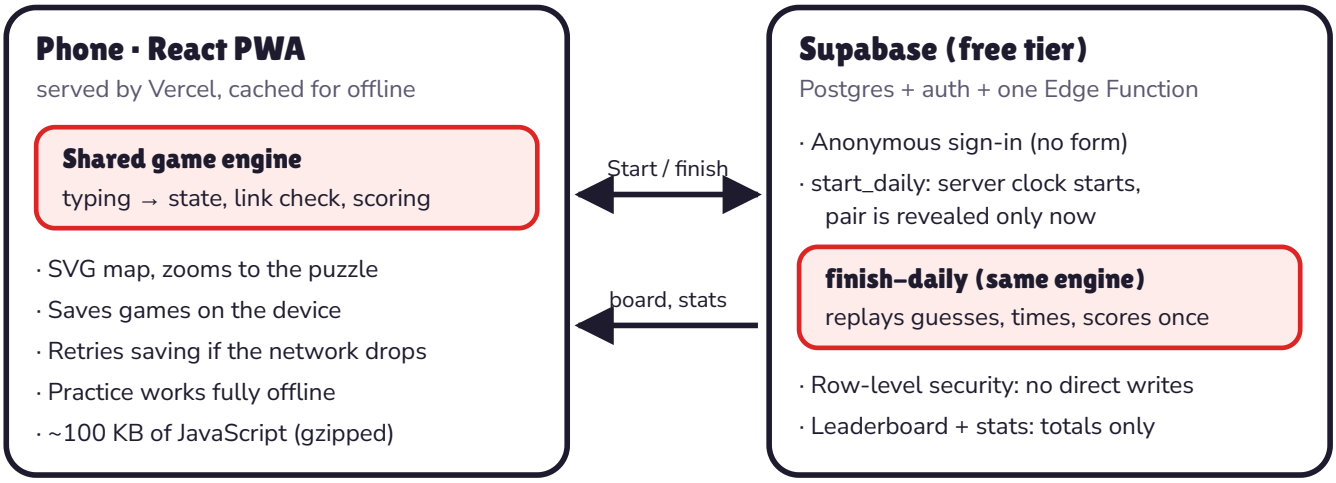
The loop is the same one that made daily word games stick: one shared puzzle, a streak to protect, and a reason to compare with others.



Mechanic	How it works	Why this way
Daily puzzle	One pair per day for everyone, picked from a year-long schedule.	A shared puzzle gives people something to talk about and compare.
Points	`100 × (k+1) × par / (par + t)` `par = 20 s + 10 s × k`	Always rewards being faster; harder pairs (bigger k) are worth more. Loss = 0.
Streak	+1 for every day you finish the daily, win or lose.	Rewards showing up. Counting losses removes any reason to type junk just to end a hard day.
Leaderboard	Today's players by time, then fewer guesses, then who finished first.	Time is the skill being measured; ties are broken fairly.
Today and You stats	Players today, median time, "faster than X%", most-guessed state; states discovered on your own map.	Gives a win-or-lose day a reason to look, and turns the map into a collection.
Practice	Random pairs, any time, works offline.	Lets people get better without farming points.

05 Tech stack and choices

The game rules live in one TypeScript engine. The phone runs it for instant feedback, and the server runs the same file again to check every result.



Layer	Choice	Why
App	React 19 + Vite + TypeScript, Tailwind	Small output, fast to build, typed end to end. Not Flutter web: its rendering engine is a heavy download for budget phones.
Offline + install	PWA (service worker, manifest)	Repeat visits load from the phone, and it can be added to the home screen like an app.
Map	Hand-built SVG from SimpleMaps data (CC BY 4.0)	No map library; the whole India map is about 21 KB gzipped. Shows India's official boundaries.
Border graph	33 playable regions, 64 borders, hand-checked	Checked edge by edge against the map, with a script that fails the build on any mismatch. Islands and Puducherry are not playable.
Puzzles	235 valid pairs, 365-day schedule, generated once	BFS proves every pair has a route of 2–5 states. The schedule is committed, so it can't drift between users.
Backend	Supabase Postgres, RLS, SQL functions, 1 Edge Function	Free tier, real database rules, and the server re-uses the exact game engine.
Hosting	Vercel + a daily GitHub Action	Free; the Action pings the database so the free project never pauses.
Testing	Vitest, Playwright (incl. iPhone WebKit), pgTAP	99 unit, 96 browser and 66 database tests, plus a 13-step smoke test on the live server.

06 Kenpath's three constraints

Modest Android phones and mobile data

≈ 210 KB first visit, everything included	**≈ 0 KB** repeat visits (cached on the phone)	**≈ 10 KB** one full daily game
2.1 s first screen on Slow 4G + 4× slower CPU	**≈ 1.2 MB** stored on the device	**0** installs or app-store downloads needed

Fonts are self-hosted and trimmed (the pixel logo font is 1 KB). The server library loads after the first screen appears. Sounds are generated in code, so there are no audio files to download.

English as a second language

- How to play is three cards with a picture and six words each: "Two states light up. Join them by land."
- Typing accepts full names, abbreviations (MP, J&K), old names (Orissa, Uttaranchal) and Hindi names (मध्य प्रदेश).
- Typos get one "Did you mean" chip; nothing is ever accepted silently. Partial names get "Type the full name", never "wrong".
- Messages are short and plain: "Already guessed", "Not linked yet", "That's your start".

Points worth real money

Risk	What stops it (tested on the live server)
Seeing the puzzle early	The pair is never in the app bundle; the server reveals it only when you press Start, and starts the clock at the same moment.
Faking a fast time	Time is server finish minus server start, not the phone's clock.
Faking a win	The server replays every guess on its own copy of the map with the same engine.
Playing the daily twice	One daily per player per day, enforced by the database.
Lost or doubled points	Saving retries until it succeeds; finishing twice returns the same stored result.
Bots	Finishes faster than 1.5 s + 1 s per guess are stored but flagged and kept off the board.
Farming points	Practice earns nothing.

What this does not cover yet: players are anonymous, so clearing browser data loses progress and one person could make several accounts; official time includes network delay; and the bot flag is a heuristic. Real payouts would need verified accounts first.

07 With more time

I kept the build to what I could finish and test properly. These are the next steps, roughly in the order I would build them.

Idea	What it adds	Effort
Proper accounts	Phone OTP login linked to the existing anonymous player, so the streak and points carry over. Needed before any real payout.	first
Language picker	Full UI in Hindi, Marathi, Tamil, Bengali and more; state names shown in the player's script.	medium
Voice in, voice out	Speak a state name (speech-to-text, confirmed before it counts) and hear the puzzle and result read aloud (text-to-speech) for players who read less.	medium
Android app	Wrap the same code as an APK (Trusted Web Activity) for the Play Store, with real push reminders before a streak breaks.	small
State leaderboards	Pick a home state (or detect it with GPS), see the top players in your state, and an inter-state league of average times.	medium
Friends and sharing	Private groups and a WhatsApp share card with an emoji route that doesn't spoil the answer.	small
Review tools	A dashboard to clear or reject flagged times, plus retention and completion analytics.	small

New ways to play

Blind mode
No map at all: connect the states from memory. Double points for the brave.

Districts
Same game inside one state, district by district. A new level for players who have mastered the states.

Blitz
As many pairs as you can in two minutes. Good for practice and short breaks.

Rivers and rails
Connect two cities along a river or a train line, and learn real geography on the way.

Weekly long route
One very long pair each Sunday, e.g. Ladakh to Mizoram, for a bigger reward.

Hints with a cost
Reveal one neighbour for +10 seconds, so stuck players keep going instead of quitting.

08 What I left out, on purpose

Left out	Why
Sign-in accounts	A form before the first game hurts a daily habit; anonymous play first, verified accounts later (above).
GPS and state leaderboards	A location prompt before playing feels intrusive; one fair India-wide board first.
Full translation, voice	Worth doing properly rather than half done; input already accepts Hindi names.
Badges, streak freezes, push	The core loop had to be solid and fair before adding more layers.

Map note: the map file predates a 2014 boundary change near Khammam. The game follows a hand-checked border list (Andhra Pradesh–Chhattisgarh linked, Odisha–Telangana not), documented in the repository.

09 Where I used AI tools

I used Claude as an AI coding assistant. I came up with the game, made the product and rule decisions, reviewed every checkpoint and tested on real phones. AI helped me stress-test the design, write much of the code and tests to my specification, run browser and database checks, and draft documentation.

10 Links

What	Where
Live app	https://sarhad-teal.vercel.app
Repository	https://github.com/nitya2405/sarhad
Setup and architecture	README
Game rules and test matrix	docs/DESIGN.md
Visual system	docs/DESIGN-SYSTEM.md
Map source and border rulings	data/SOURCES.md · data/ADJACENCY_NOTES.md



Hope you like it :)

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